Day 2: DHCP Configuration

Task 1 : In Cisco Packet Tracer, set up a simple network with one DHCP server and two client PCs. Configure the DHCP server to assign IP addresses automatically, then verify on each PC that it receives a different IP address from the DHCP pool.

To configure a DHCP server in Cisco Packet Tracer and automatically assign IP addresses to two client PCs.

Devices Used:

- 1 Server-PT
- 1 Cisco 2960 Switch
- 2 PC-PT
- Copper Straight-Through cables

DHCP Server Configuration:

Setting	Value
Server IP	192.168.1.2
Subnet Mask	255.255.255.0
Pool Name	LAN
Start IP Address	192.168.1.100
DNS Server	8.8.8.8
Maximum Users	10

Procedure:

1. Created a network topology using one server, one switch, and two PCs.
2. Connected all devices using Copper Straight-Through cables.
3. Configured a static IP address on the server.
4. Enabled DHCP service from the Server's Services tab.
5. Created a DHCP pool with the starting IP address 192.168.1.100.
6. Selected DHCP in the IP Configuration window of both PCs.
7. Verified that both PCs automatically received IP addresses from the DHCP server.

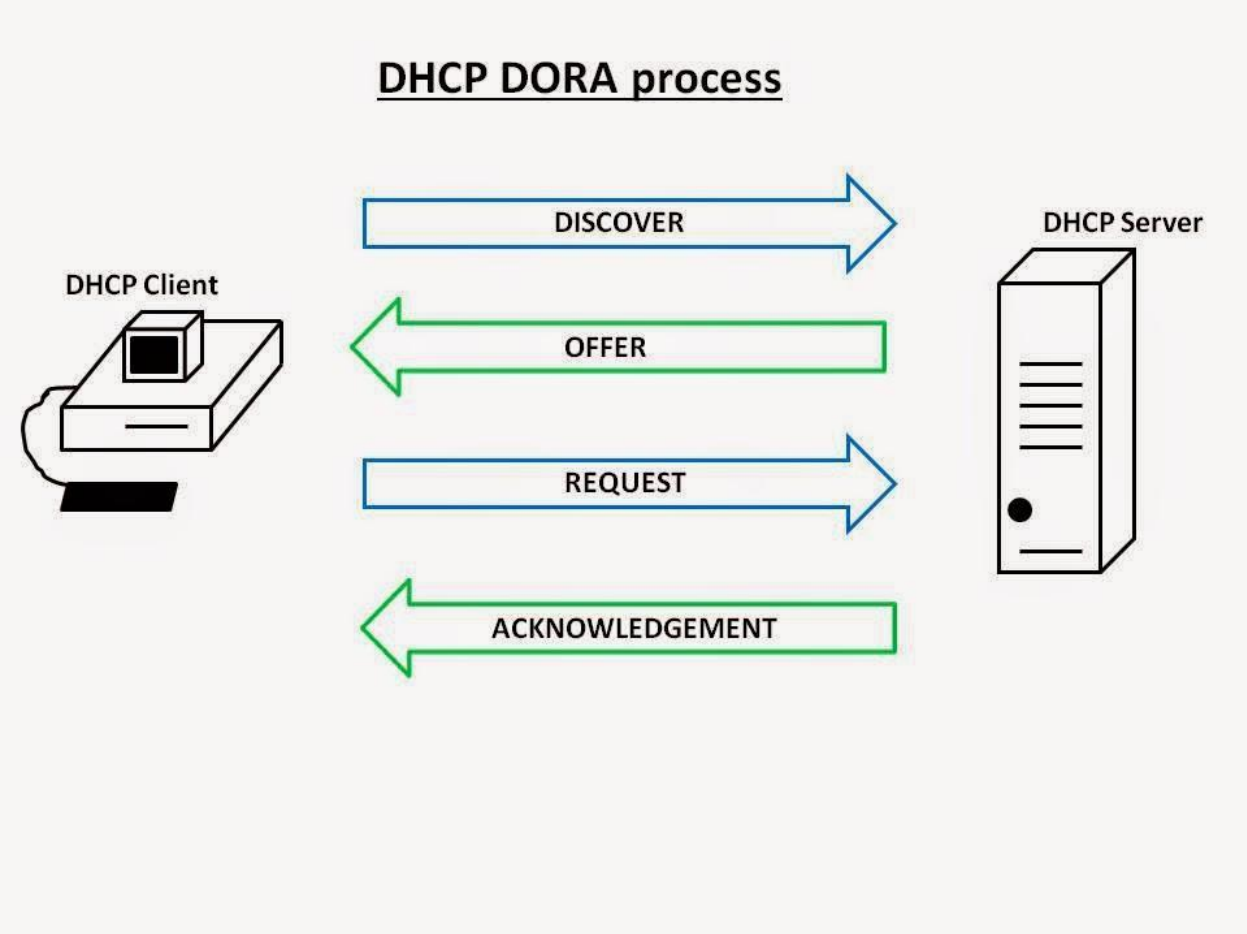
Observed Result:

Device	IP Address
Server	192.168.1.2
PC0	192.168.1.100
PC1	192.168.1.101

Result: Successfully configured a DHCP server and assigned different IP addresses automatically to two client PCs.

Conclusion: DHCP simplifies network configuration by automatically assigning IP addresses and other network settings to connected devices.

Task 2 : Draw a sequence diagram (hand-drawn or digital) that clearly shows each step of the DHCP process: Discover, Offer, Request, and Acknowledge, labeling the sender and receiver at each stage.



Explanation of DHCP Process (DORA)

DHCP uses a four-step process called DORA to automatically assign IP addresses to client devices.

1. **Discover:** The client broadcasts a DHCP Discover message to find an available DHCP server.
2. **Offer:** The DHCP server sends a DHCP Offer message containing an available IP address and network configuration details.
3. **Request:** The client sends a DHCP Request message to indicate that it wants to use the offered IP address.
4. **Acknowledge:** The DHCP server sends a DHCP Acknowledgement (ACK) message to confirm the IP address assignment.

Task 3 : Change the DHCP pool range on your Cisco Packet Tracer DHCP server so that only one IP address is available. Add two PCs and observe what happens when both try to get an IP address. Take a screenshot of the result and explain in 2-3 lines why only one PC succeeds.

Procedure:

1. Opened the DHCP server in Cisco Packet Tracer.
2. Changed the DHCP pool configuration to allow only one IP address.
3. Set the starting IP address to 192.168.1.100.
4. Connected two client PCs to the network.
5. Selected DHCP in the IP Configuration window of both PCs.
6. Observed the IP address assignment results.

DHCP Pool Configuration:

Setting	**Value**
Pool Name	LAN
Start IP Address	192.168.1.100
Subnet Mask	255.255.255.0
Maximum Users	1

Observed Result:

Device	**Result**
PC0	192.168.1.100 assigned successfully
PC1	DHCP request failed because no IP address was available

Explanation:

The DHCP pool contains only one available IP address. PC0 receives the available IP address successfully, but PC1 cannot obtain an IP address because the only available address has already been assigned to PC0.

Result:
Successfully tested DHCP address allocation with a limited IP address pool.

Task 4 : Simulate a Zomato-style restaurant WiFi scenario in Cisco Packet Tracer: configure a DHCP server to provide IP addresses to both a laptop and a smartphone (use a generic PC as smartphone). Show the assigned IPs and explain why DHCP is useful in this situation.

To configure a restaurant-style wireless network where a laptop and a smartphone receive IP addresses automatically using DHCP.

Devices Used:

- 1 Server-PT
- 1 Wireless Router (WRT300N)
- 1 Laptop-PT
- 1 PC-PT (simulated smartphone)

Network Configuration:

Setting	**Value**
Router IP Address	192.168.10.1
Server IP Address	192.168.10.2
DHCP Pool Name	Restaurant
DHCP Start IP	192.168.10.100
Subnet Mask	255.255.255.0
DNS Server	8.8.8.8
Maximum Users	20
Wi-Fi SSID	Zomato_Restaurant

Procedure:

1. Created a wireless network using a WRT300N router.
2. Connected the DHCP server to the router using an Ethernet cable.
3. Configured a static IP address on the server.
4. Enabled DHCP service and created a pool named Restaurant.

5. Configured the wireless network with the SSID Zomato_Restaurant and WPA2 security.
6. Connected the laptop and PC (simulated smartphone) to the restaurant Wi-Fi.
7. Selected DHCP on both client devices to obtain IP addresses automatically.

Observed Result:

Device	Assigned IP Address
Server	192.168.10.2
PC (Smartphone)	192.168.10.100
Laptop	192.168.10.101

Why is DHCP Useful in This Scenario?

In a restaurant, many customers connect their devices to the same Wi-Fi network. DHCP automatically assigns IP addresses and other network settings to each device without requiring manual configuration.

It saves time, reduces configuration errors, and makes network management easier.

Result:
Successfully configured a restaurant-style wireless network and provided automatic IP address assignment to both client devices.

Conclusion:
DHCP makes it easy for multiple customers to connect their smartphones and laptops to a restaurant Wi-Fi network without manually configuring IP addresses.